DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

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MITRE ATT&CK Matrices

1. What is a MITRE ATT&CK Matrix?

A MITRE ATT&CK matrix is a way of organizing attacker behaviour into different **tactics and techniques**.

It helps security teams understand how attackers behave in different environments.

Different environments have different types of systems and attacks, so MITRE has separate matrices for them.

The three main matrices I learned about are:

- Enterprise
- Mobile
- ICS (Industrial Control Systems)

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2. Enterprise Matrix

The **Enterprise Matrix** is used for attacks against normal enterprise environments.

It mainly covers environments such as:

- Windows systems
- Linux systems
- macOS
- Cloud environments
- Network infrastructure
- Enterprise applications

It contains tactics and techniques that attackers commonly use when targeting organizations.

Example

An attacker sends a phishing email to an employee.

- **Tactic:** Initial Access
- **Technique:** Phishing

If the attacker then uses PowerShell after getting access:

- **Tactic:** Execution
- **Technique:** Command and Scripting Interpreter: PowerShell

The Enterprise Matrix is especially useful for SOC analysts because most SOC investigations involve enterprise systems, endpoints, servers, networks and cloud environments.

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3. Mobile Matrix

The **Mobile Matrix** focuses on attacks against mobile devices and mobile operating systems.

It mainly covers platforms such as:

- Android
- iOS

Attackers can use different methods on mobile devices compared to normal computers, so the Mobile Matrix has techniques related to mobile environments.

Example

An attacker may try to distribute a malicious application or abuse a mobile device to collect information.

The Mobile Matrix helps security teams understand these types of attacks and identify the techniques being used.

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4. ICS Matrix

ICS stands for **Industrial Control Systems**.

The ICS Matrix is designed for environments where industrial systems and control equipment are used.

Examples include:

- Power plants
- Manufacturing systems
- Water treatment facilities
- Industrial control environments
- Critical infrastructure

These environments are different from normal office networks because an attack can potentially affect physical processes and equipment.

Example

An attacker may try to gain access to an industrial control system and then manipulate a process or affect the availability of the system.

The ICS Matrix helps security teams understand the techniques attackers may use against these environments.

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5. Comparison of the Three Matrices

Matrix	Main Environment	Used For
Enterprise	Organizations, computers, servers, cloud and networks	Enterprise cybersecurity and SOC investigations
Mobile	Android and iOS devices	Mobile security and mobile attack investigation
ICS	Industrial Control Systems	Industrial and critical infrastructure security

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6. Why are different matrices needed?

I understood that the same attack method cannot always be applied to every environment.

For example, a Windows enterprise computer and an industrial control system work very differently.

A SOC analyst investigating a Windows endpoint may look at things like:

- PowerShell
- Windows Event Logs
- Processes
- User accounts
- Network connections

But an analyst working with an industrial environment may need to look at industrial protocols, control systems and physical processes.

That's why MITRE separates the techniques into different matrices.

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7. My Understanding

The main thing I learned is that the MITRE ATT&CK Matrix is basically a structured way to view attacker behaviour.

Enterprise is mainly for normal organizational IT environments, **Mobile** is for mobile devices, and **ICS** is for industrial control environments.

As someone learning SOC analysis, the **Enterprise Matrix** is the one I will probably use the most because SOC analysts commonly investigate Windows/Linux endpoints, servers, networks and cloud environments.

Knowing which matrix applies to the environment is important because it helps the analyst look for the right type of attacker behaviour.